

MaxPool1d#

<https://pytorch.org/docs/stable/generated/torch.nn.MaxPool1d.html>

Description:

Applies a 1D max pooling over an input signal composed of several input planes. In the simplest case, the output value of the layer with input size (N, C, L) and output (N, C, L_{out}) can be precisely described as: If padding is non-zero, then the input is implicitly padded with negative infinity on both sides for padding number of points. dilation is the stride between the elements within the sliding window. This link has a nice visualization of the pooling parameters. When `ceil_mode=True`, sliding windows are allowed to go off-bounds if they start within the left padding or the input. Sliding windows that would start in the right padded region are ignored.

Function Signature

```
class torch.nn.MaxPool1d(kernel_size, stride=None,  
padding=0, dilation=1, return_indices=False,  
ceil_mode=False)[source]#
```

Parameters

Shape:

```
forward(input)[source]#
```

Code Examples

```
>>> # pool of size=3, stride=2  
>>> m = nn.MaxPool1d(3, stride=2)  
>>> input = torch.randn(20, 16, 50)  
>>> output = m(input)
```

Notes & Warnings

NOTE

When `ceil_mode=True`, sliding windows are allowed to go off-bounds if they start within the left padding or the input. Sliding windows that would start in the right padded region are ignored.

Detailed Documentation

Documentation

Applies a 1D max pooling over an input signal composed of several input planes.

In the simplest case, the output value of the layer with input size (N, C, L) and output (N, C, L_{out}) can be precisely described as:

If padding is non-zero, then the input is implicitly padded with negative infinity on both sides for padding number of points. dilation is the stride between the elements within the sliding window. This link has a nice visualization of the pooling parameters.

When `ceil_mode=True`, sliding windows are allowed to go off-bounds if they start within the left padding or the input. Sliding windows that would start in the right padded region are ignored.

`kernel_size (Union[int, tuple[int]])` – The size of the sliding window, must be > 0 .

`stride (Union[int, tuple[int]])` – The stride of the sliding window, must be > 0 . Default value is `kernel_size`.

`padding (Union[int, tuple[int]])` – Implicit negative infinity padding to be added on both sides, must be ≥ 0 and $\leq \text{kernel_size} / 2$.

`dilation (Union[int, tuple[int]])` – The stride between elements within a sliding window, must be > 0 .

`return_indices (bool)` – If True, will return the argmax along with the max values. Useful for `torch.nn.MaxUnpool1d` later

`ceil_mode` (bool) – If True, will use ceil instead of floor to compute the output shape. This ensures that every element in the input tensor is covered by a sliding window.

Input: (N, C, L_{in}) or (N, C, L_{in}) or (C, L_{in}) or (C, L_{in}) .

Output: (N, C, L_{out}) or (N, C, L_{out}) or (C, L_{out}) or (C, L_{out}) ,

where `ceil_mode = False`

where `ceil_mode = True`

Ensure that the last pooling starts inside the image, make $L_{out} = L_{out} - 1$ when $(L_{out} - 1) * \text{stride} \geq L_{in} + \text{padding}$.

Runs the forward pass.